Assignment 3 - Deliver

Due 12th January 2027

OVERVIEW AND GOALS

In this third and final exercise your task is to demonstrate what you have achieved in the previous few months. You will do this in three different forms:

- Build a small demo application that runs your trained model and performs inference
- Write the final report of your project
- Prepare a presentation that you'll give in class

DELIVERABLES

Demo Application

Even the best scientific research is worthless, if you can't share your knowledge and communicate your results. Therefore, you are asked to build a small demo application that a potential user could run to understand what you have done. There are many ways you can realize this. In its simplest form, it is a command-line tool that takes the user input and produces a file as a result. A more sophisticated way would be to have a docker container that wraps such a command-line tool and could be deployed to a server for others to use. But depending on your project, a completely different form might be appropriate.

A great way to build a sample application is to develop a simple JavaScript web application that runs your trained model in the browser and provides a minimal interface to trigger inference. Some tools for doing this are Streamlit, Gradio, ONNX Runtime, Tensorflow.js, Anvil, and Dash. You can upload your trained models to Github (as an attachment to a release - don't commit it into the repository!) and dynamically fetch them when the user opens the web application.

As a reference, you can play with the Deep Optical Measure Detector, which was developed for recognizing the structure (measures) in music scores. It is a Vue.js application and available on Github.

Final Report

Your final report should contain a concise and informative written summary of your project. A large section of that report can be taken from your project proposal with appropriate edits. The entire report should be **no longer than 5 pages**.

It should answer the four fundamental questions:

- What is the problem that you tried to solve?
- Why is it a problem?
- What is your solution?
- Why is it a solution? (And in particular, why is or isn't deep learning a solution?)

Additionally, it should cover:

- The main take-aways and insights you gained from your project (e.g., batch-normalization improved the results significantly, Adadelta worked much better than SGD on my data, annotating data works really good with tool X, setting up the pre-processing takes much more time than I expected, ...)
- If you would do the same project again, what - if anything - would you do differently?
- How much time did you spend on your project? How does the number compare to your initial estimate? If you underestimated any part, what were the reasons for this?

The final report should be submitted as a single PDF file.

Presentation

Prepare a presentation that you will give in the last three classes. The presentation **must not be longer than four minutes**. The first two minutes should be dedicated to a quick introduction into your topic, the selected approach and the final results. The remaining two minutes should focus on insights that you want to share with your colleagues and ideally a small demo of the project.

Please reserve the full 2 hours for the last three lectures. For fairness, all submissions must be handed in before the first of these three lectures and the actual presenters will be selected randomly.

The presentation must be added to the Master slide deck of one of the following three online tools: Canva, Google Slides, Powerpoint Online. If you want to demo your project, please record a screencast and embed that video into your presentation. In case you don't want to use any of these three tools, you may also submit a static PDF file. No other submission will be accepted.
Be mindful with the master slide decks and only edit your own slides.

Submission

Once you're done, please notify us by writing an e-mail to applied-deep-learning@outlook.com with the subject **[Applied Deep Learning] Assignment 3 - Your Matr.Number.** Please attach the final report (and optionally the presentation PDF in case you haven't added it to the online decks).

CRITERIA FOR GRADING

As discussed in the preliminary lecture, every assignment will be graded according to these five criteria (as applicable):

- Results
- Creativity
- Complexity
- Code Quality
- Presentation

You may achieve up to 26 points in this assignment with:

- the demo application being worth up to 8 points
- the report being worth up to 10 points
- the presentation being up to 8 points

Late Policy

Submissions will still be accepted after the deadline, but keep in mind that there is a penalty of one point per day. This applies to the demo application and the report.

Please note that the presentation is exempt from this policy, as you have to present live. If you are absent, this part will be graded with 0 points. If you know that you can't make one of the three dates, please e-mail us as soon as possible and we'll make arrangements.

OTHER QUESTIONS

In case you have other questions regarding the assignment, please send an e-mail to applied-deep-learning@outlook.com or post your question in the discussion forum in ILEA (preferred).